



Smart Agriculture Pest Detection

A Mist-Fog-Assisted Real-Time Framework Using Deep Transfer Learning

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Introduction

Revolutionizing Farming with Smart Agriculture 4.0

The Agricultural Crisis



Global Yield Loss

Pests and pathogens account for nearly 30% of annual crop yield losses globally, threatening food security.



Bandwidth Barriers

High-resolution visual data creates "data storms" that saturate rural networks in purely cloud-centric models.



The Latency Wall

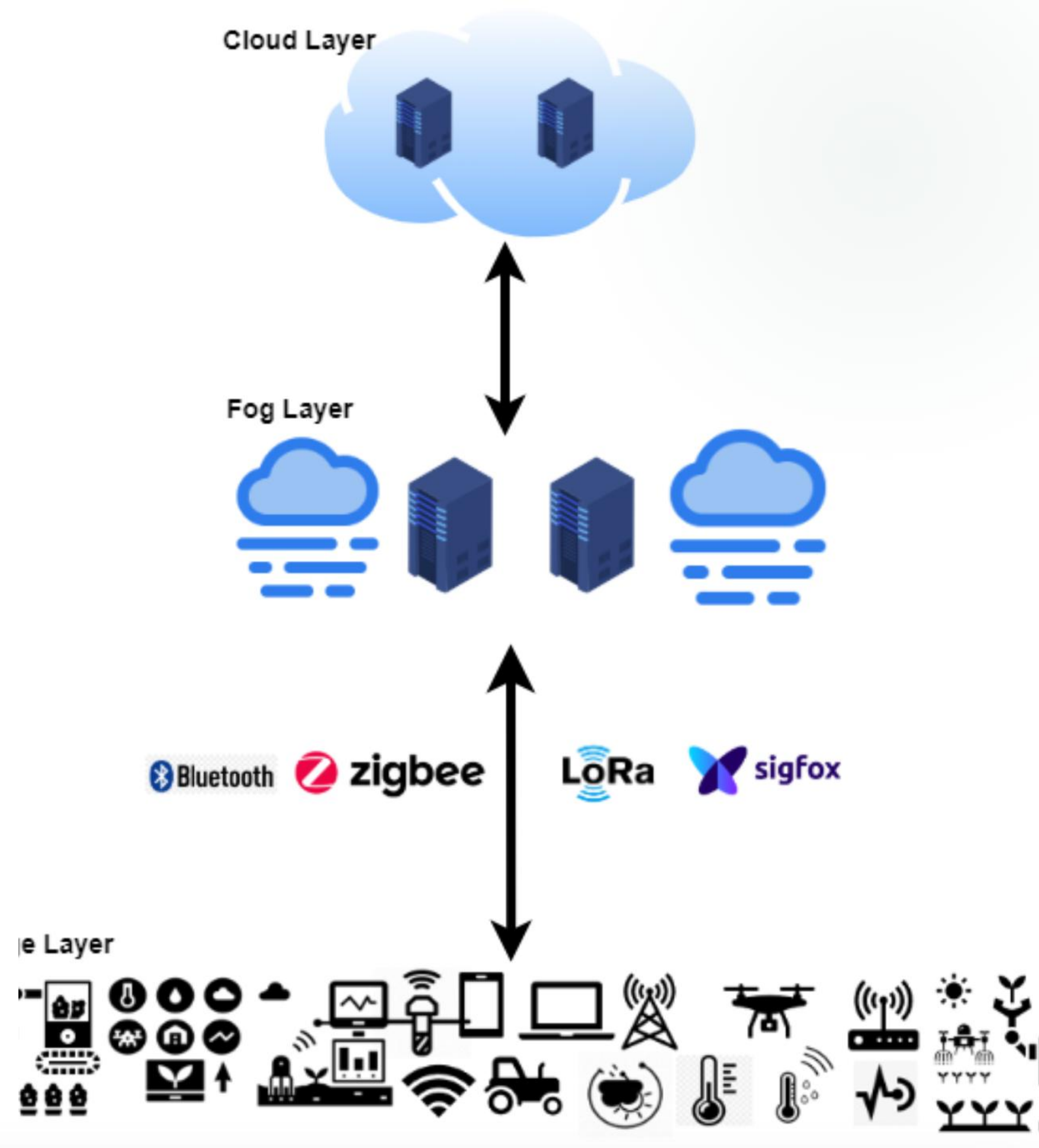
Real-time detection requires localized processing to ensure immediate actuation (e.g., localized spraying).

System Architecture

Mist-Fog-Cloud Hierarchy

Our decentralized framework distributes the computational load across three specialized layers to ensure responsiveness and stability.

By moving "Cognitive" tasks to the Fog layer and "Reflexive" tasks to the Mist layer, we optimize both speed and accuracy.



Distributed Intelligence



Mist Node

Raspberry Pi 4

Handles image acquisition, noise reduction, and initial TFLite detection.



Fog Node

Local Server

Performs high-fidelity classification using full DINOv2-ViT model (15ms latency).

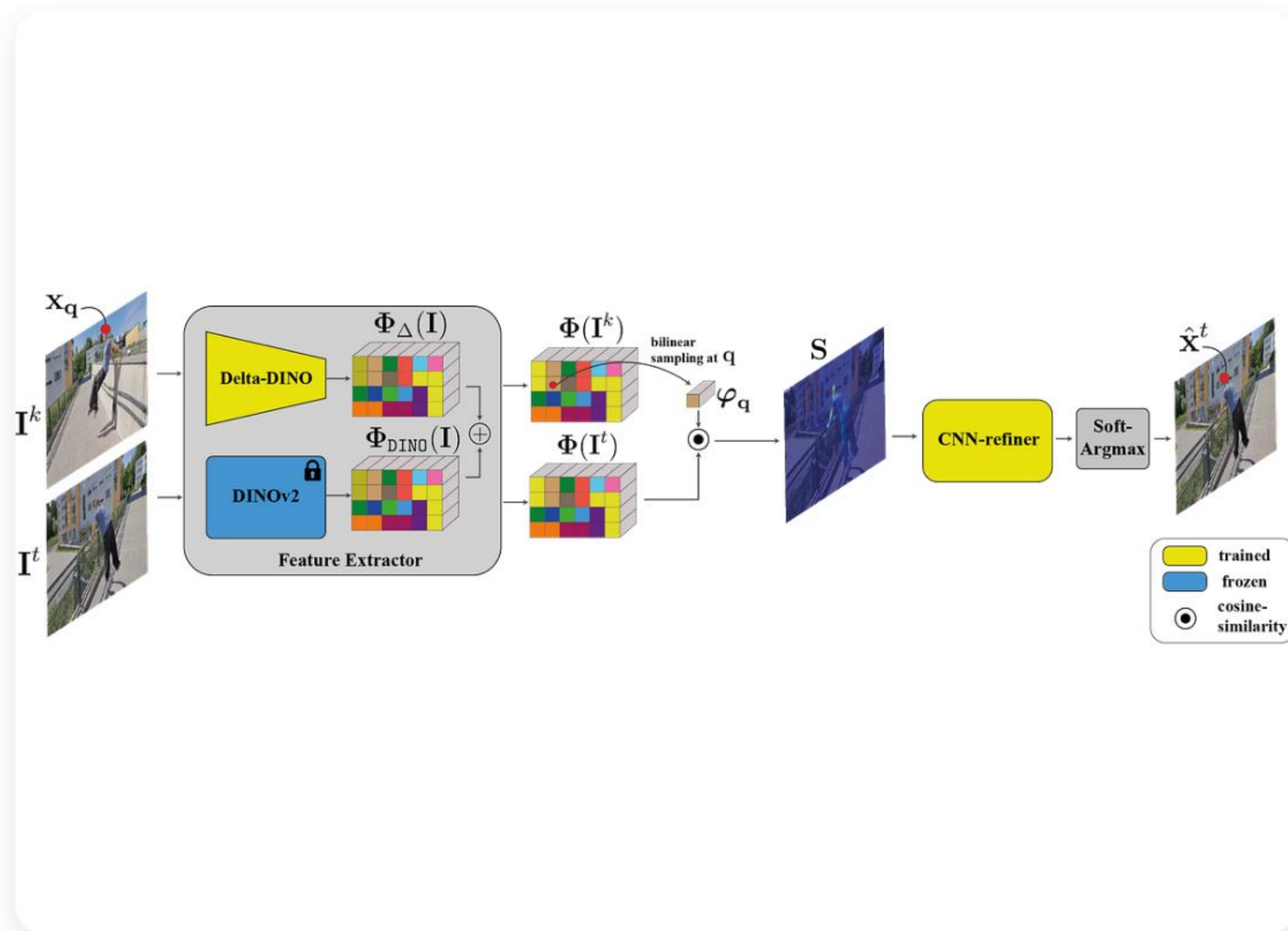


Cloud Tier

Remote Analytics

Manages long-term storage, migration pattern mapping, and model retraining.

DINOv2: The Vision Transformer



Self-Supervised Feature Extraction

- ✓ **vits14 Backbone:** Segments images into 14x14 pixel patches for global context.
- ✓ **Fine-Grained Classification:** Distinguishes 132 pest categories in the Pestopia dataset.
- ✓ **Label Smoothing (0.1):** Improves generalization on unseen test samples.

Intelligent Offloading Logic

32%

Offloaded Frames

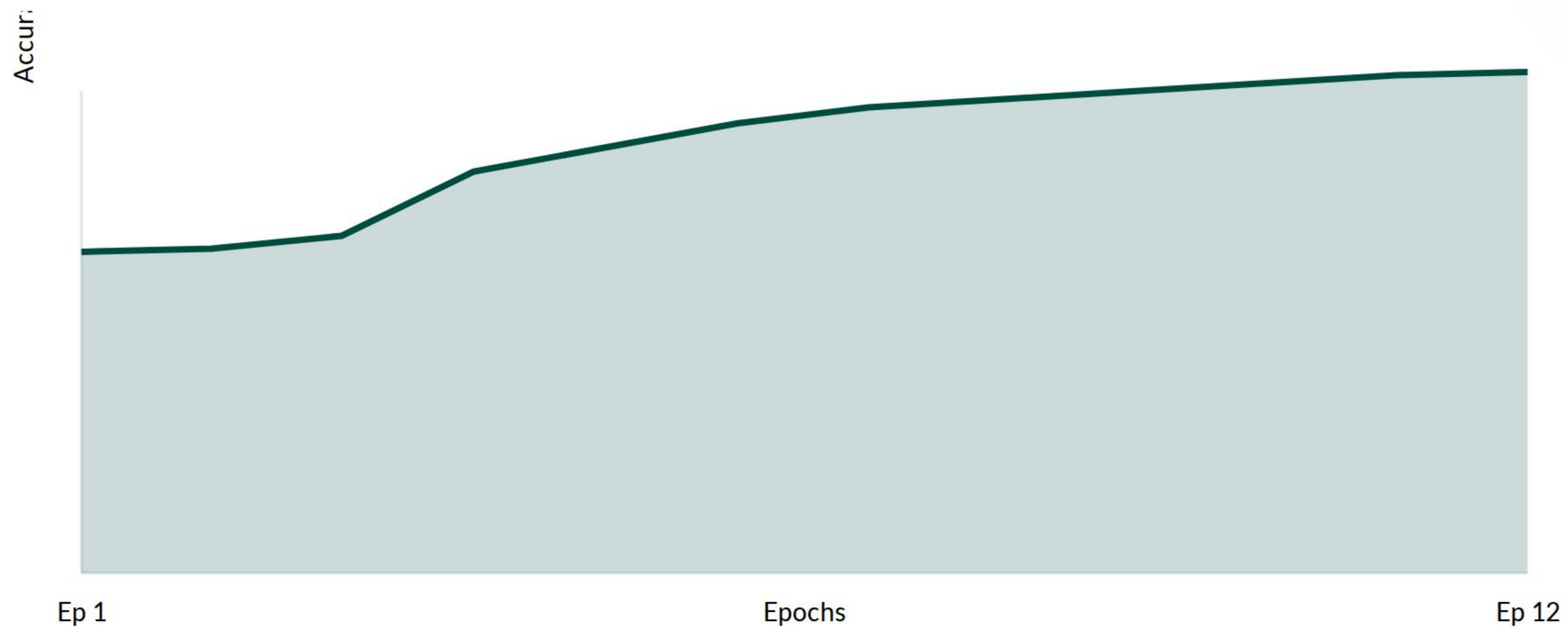
M/M/1/K Queuing Theory

To prevent system overflows during pest "bursts", we employ Markov Modulated Poisson Process (MMPP).

$$P_{\text{offload}} = \max\left(0, \frac{\lambda}{\mu + \lambda (k - 1)}\right)$$

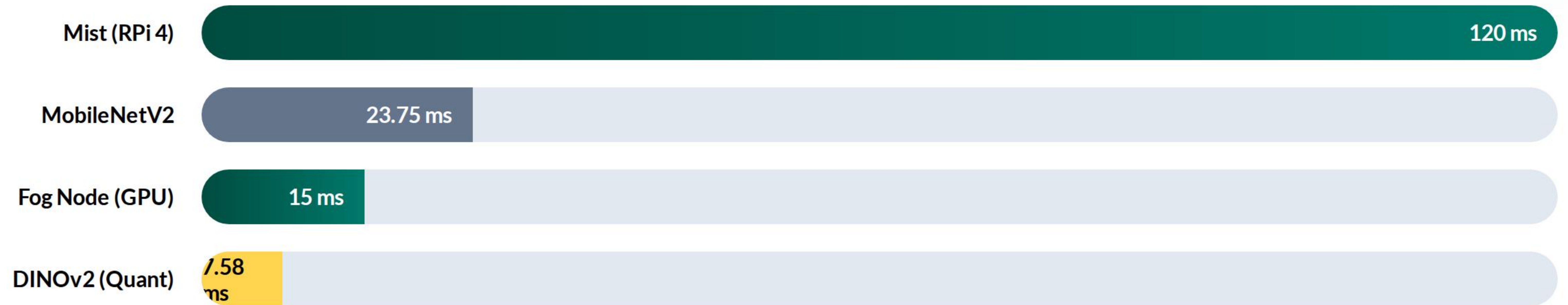
This dynamic calculation ensures a stable monitoring rate of 10 FPS.

Training Accuracy (Pestopia Dataset)



Validation accuracy peaked at 76.5% following the Stage 2 Fine-tuning transition.

Latency Comparison (ms)



Fog-layer processing provides an 8x speedup over localized Raspberry Pi inference.

Model Performance Metrics

Metric	MobileNetV2 (Baseline)	DINOv2-S (Proposed)	Improvement
Top-1 Accuracy	63.4%	72.1%	+13.7%
Macro F1-Score	0.61	0.71	+16.4%
Inference Latency	23.75 ms	7.58 ms (Quantized)	+68.1%
Model Size	37.3 MB	12.23 MB	+67.2%

Conclusion & Future Directions

The Next Frontier

- ✓ **Blockchain Integration:** Ensuring data sovereignty and a verifiable audit trail for pest outbreaks.
- ✓ **Mobility-Aware AI:** Adaptive offloading for autonomous tractors and UAV swarms.

Summary

Decentralized computing successfully solves the "Bandwidth Wall" problem in rural IoT.

The DINOv2 backbone achieves high-resolution understanding of 132 pest species with superior efficiency.

Thank You!

Questions & Discussion

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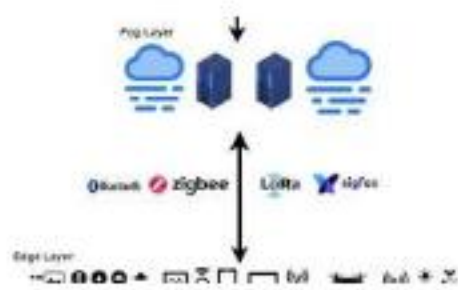
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Image Sources



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